MEDINOVA TEACHING HOSPITAL

Class 03 — Acute Emergencies

STEMI · Respiratory Failure · Anaphylaxis · Septic Shock

> *"In the acute setting, the first five minutes determine the next five hours."*

01. The ABCDE Approach

STANDARD PROTOCOL

The **ABCDE approach** is the gold standard for assessing and treating the critically ill patient. Treat life-threatening problems as you find them before moving to the next step.

Step	Assessment (Look/Listen/Feel)	Interventions
A — Airway	Patency, stridor (noisy breathing), gurgling.	Head tilt/Chin lift, Oropharyngeal airway (OPA), Suction.
B — Breathing	SpO2, Resp Rate, Chest expansion, Auscultation.	High-flow Oxygen (15L Reservoir mask), Nebulisers.
C — Circulation	BP, Heart Rate, CRT (Capillary Refill Time), Urine output.	IV Access (x2 large bore), IV Fluid bolus (500ml saline).
D — Disability	GCS (Glasgow Coma Scale), Pupils, Glucose (BM).	Recovery position, IV Glucose if hypoglycaemic.
E — Exposure	Full body check, temperature, rashes, calves (DVT).	Keep patient warm, maintain dignity.

> **Prof. Obi's Pearl #1**
>
> *"The airway is a binary state: it is either protected or it isn't. If the patient can't tell you their name clearly, assume the airway is at risk."*

02. STEMI Recognition & Management

CARDIOVASCULAR EMERGENCY

STEMI: ST-Elevation Myocardial Infarction (A major heart attack caused by a complete blockage of a coronary artery).

ECG Criteria: ST elevation >1mm in 2 limb leads or >2mm in 2 chest leads, or New Left Bundle Branch Block (LBBB).

The Management Pathway

- **M.O.N.A:** Morphine, Oxygen (if SpO2 <94%), Nitrates (GTN), Aspirin (300mg).
- **DAPT:** Dual Antiplatelet Therapy (Aspirin + Ticagrelor or Clopidogrel).
- **Primary PCI:** Percutaneous Coronary Intervention (Stenting). *Door-to-balloon time goal: < 90 minutes.*
- **Thrombolysis:** "Clot-busting" drugs (e.g., Alteplase) if PCI is not available within 120 minutes.

STEMI Mimics (Beware!)

Pericarditis (Global ST elevation), *Brugada Syndrome*, or *Early Repolarization*.

03. Acute Severe Asthma

RESPIRATORY EMERGENCY

Classification	Clinical Features	Action
Moderate	PEFR 50-75% best/predicted.	SABA via spacer.
Severe	PEFR 33-50%, RR >25, HR >110. Cannot finish sentences.	Nebulised Salbutamol + Ipratropium.
Life-Threatening	PEFR <33%, SpO2 <92%, Silent chest, Cyanosis, Exhaustion.	Senior help! IV Magnesium, consider ITU.

04. Anaphylaxis

IMMUNOLOGICAL EMERGENCY

A severe, life-threatening systemic hypersensitivity reaction.

Treatment: IM Adrenaline (1:1000)
- Adult: 500 micrograms (0.5ml)
- Child (>12y): 500 micrograms (0.5ml)
- Child (6-12y): 300 micrograms (0.3ml)

Biphasic Reaction: Symptoms can return 1-72 hours after initial recovery. Patients must be observed for 6-12 hours.

> **Prof. Obi's Pearl #2**
> *"If you're debating whether to give adrenaline or not, you've already waited too long. Give it."*

05. Septic Shock & Sepsis Six

INFECTIOUS EMERGENCY

Sepsis-3 Definition: Life-threatening organ dysfunction caused by a dysregulated host response to infection.

THE SEPSIS SIX (COMPLETE WITHIN 60 MINS)

Give (3):

1. High-flow Oxygen.
2. IV Antibiotics (Broad spectrum).
3. IV Fluid Resuscitation.

Take (3):

1. Blood Cultures.
2. Lactate (Serum).
3. Urine Output monitoring.

06. Code Blue: ALS Algorithm

CARDIAC ARREST

4Hs & 4Ts (Reversible Causes):

Hypoxia **H**ypovolaemia **H**yper/Hypokalaemia **H**ypothermia	**T**ension Pneumothorax **T**amponade (Cardiac) **T**oxins **T**hrombosis (PE/MI)

> **Prof. Obi's Pearl #3**
> *"During a Code Blue, the loudest person in the room should be the timer. Chaos is the enemy of resuscitation."*

07. Emergency Drug Reference

QUICK ACCESS TABLE

Drug	Standard Adult Dose	Indications
Adrenaline	1mg (1:10,000 IV) every 3-5 mins	Cardiac Arrest
Amiodarone	300mg IV Bolus	Refractory VF / Pulseless VT
Atropine	500 micrograms IV	Symptomatic Bradycardia
Adenosine	6mg -> 12mg -> 12mg (Rapid IV push)	SVRT / Tachycardia diagnosis
Magnesium	2g IV over 10-20 mins	Torsades de Pointes / Severe Asthma

> **Prof. Obi's Pearl #4**
> *"The first pulse you check in a crash call is your own. Slow down, breathe, and follow the algorithm. It's there for a reason."*

Medical Definitions:
- **Thrombolysis:** Breaking down blood clots using pharmacological agents.
- **Cyanosis:** Bluish discoloration of skin due to low oxygen levels.
- **ROSC:** Return of Spontaneous Circulation (the heart starts beating again).
- **DVT:** Deep Vein Thrombosis (blood clot in the leg).

OFFICIAL SIGN-OFF

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